

Challenge (Habanero)

- Q1.** If $d = rt$, and you want to isolate r , what is the new equation?
(A) $r = \frac{d}{t}$
(B) $r = dt$
(C) $r = \frac{t}{d}$
(D) $r = t - d$
(E) $r = d + t$
- Q2.** Solve for x : $2x + 5 = 11$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$
- Q3.** A hiker travels 15 miles in 3 hours. If $d = rt$, and $d = 15$, $t = 3$, what is r ?
(A) $r = 5$
(B) $r = 6$
(C) $r = 4$
(D) $r = 3$
(E) $r = 2$
- Q4.** If $a = \frac{b}{c}$, what is b ?
(A) $b = ac$
(B) $b = \frac{a}{c}$
(C) $b = a - c$
(D) $b = a + c$
(E) $b = \frac{c}{a}$
- Q5.** Solve for y : $y - 2 = 7$
(A) $y = 5$
(B) $y = 6$
(C) $y = 7$
(D) $y = 8$
(E) $y = 9$
- Q6.** A car's fuel consumption is given by $c = \frac{d}{m}$. If $c = 25$, $d = 200$, what is m ?
(A) $m = 8$
(B) $m = 10$
(C) $m = 12$
(D) $m = 6$
(E) $m = 4$
- Q7.** Solve for x : $x + 2 = 9$
(A) $x = 5$
(B) $x = 6$
(C) $x = 7$
(D) $x = 8$
(E) $x = 11$
- Q8.** If $p = \frac{r}{t}$, and $p = 4$, $r = 20$, what is t ?
(A) $t = 4$
(B) $t = 5$
(C) $t = 6$
(D) $t = 8$
(E) $t = 10$

Open Ended Questions (FRQ)

- Q9.** Solve for z : $z - 3 = 2z + 5$. Show all work.
- Q10.** If $a = \frac{b+c}{2}$, and $a = 6$, $b = 10$, what is c ? Show all work.

Chef's Tips

- Emphasize the importance of isolating variables
- Provide examples of real-world applications
- Encourage students to check their work